ASSESSMENT AND INTERNAL VERIFICATION FRONT SHEET (Individual Criteria)

(Note : This version is to be used only for assignments uploaded via Classter)

Course Title	Advanced Diploma in Software Development	Lecturer Name & Surname	
Unit Number & Title	ITSFT-406-1508: Server-side Scripting		
Assignment Number, Title / Type	Synoptic assignment: Plan and Design a web application using PHP and MySQL		
Date Set	13/06/2025	Deadline Date	20/06/2025

Student Name		ID Number		Class / Group	

Assessment Criteria	Mark
LO1 – Explain the client/server architecture employed by the World Wide Web, with reference to the stateless nature of the web and how developers work around this architecture	
KU 1.1 - Describe the client/server model with reference to web applications.	*10*
KU1.2 - Identify the components of a web application	*10*
AA1.3 – Compare and contrast static and dynamic web pages	*15*
AA1.5 – Distinguish between stateful and stateless web applications and identify techniques for tackling the stateless web	*10*
L02 - Design the layout and formatting of a web application, as well as user-interaction elements such as links, tables and forms using HTML, CSS and JavaScript to given specifications.	
KU 2.1 - List HTML tags to build static and interactive web pages.	10
KU2.3 - Describe how to use CSS selectors to style and format HTML pages, and how to use JavaScript to enhance pages.	10
LO3 – Use a serve-side scripting language in conjunction with a front-end web application layout to create a fully functional web application to given specifications	
SE3.4 – Device code that implements HTTP request handling including sessions, cookies, files and mail.	*20*
LO4 – Design the structure of a basic database and connect a database to a web application for data storage	
AA4.2 - Use SQL to create and modify databases	*15*
TOTAL	**100**

Notes to Students:
• This assignment brief has been approved and released by the Internal Verifier through Classter. • Assessment marks and feedback by the lecturer will be available online via Classter (Http://mcast.classter.com) following release by the Internal Verifier • Students submitting their assignment on Moodle will be requested to confirm online the following statements: **Student's declaration prior to handing-in of assignment** ❖ I certify that the work submitted for this assignment is my own and that I have read and understood the respective Plagiarism Policy **Student's declaration on assessment special arrangements** ❖ I certify that adequate support was given to me during the assignment through the Institute and/or the Inclusive Education Unit. ❖ I declare that I refused the special support offered by the Institute.

UNIT ITSFT-406-1508– SERVERSIDE SCRIPTING

ASSIGNMENT: PLAN AND DESIGN A WEB APPLICATION USING PHP AND MySQL

Unit Level: MQF LEVEL 4

General Guidelines

- This is a Take Home Assignment.
- This is an open book assignment.
- Students must carefully read the tasks and the rubrics before they attempt the answers.
- Deadline of this assignment is the **20th June 2025**
- Plagiarism is strictly prohibited.
- **Following assignment submission, the lecturer will hold submission reviews. Students will be expected to explain their code. The lecturer may decide not to award some or all marks if not satisfied that the work presented is the student's own.**
- Milestone submissions and the final assignments are to be submitted electronically on Moodle. No hard copies are to be submitted.
- **Lecturer can deduct marks if student does not implement all features as required per specification in the assignment. E.g. if client-side validation is performed on only one form when the assignment requires to have at least 2 forms (update form and adding form) marks will be deducted accordingly.**

Submit milestones as specified in the table below.

Milestone	Task	Date	Deliverable
1	Final submission	20th June	All sections

SECTION A - Proposal Document Template

Choose a scenario from your web application from the ones given below. Note that each one includes an example as a guide. The lecturer will in turn give you feedback and discuss with you in case any changes are necessary. **Kindly keep with the proposal deadline as specified by your lecturer.**

- **Sports**: Create a website where users can view and update news for their favourite sports teams.
- **Transport (Not Rental)**: Build a platform that displays public transport schedules and allows users to search for routes.
- **Music**: Develop a simple music library where users can browse and play songs from different genres.
- **Selling of goods or services (Not Rental)**: Design an online shop where users can add products to a cart and place orders.
- **Beauty & Fashion**: Set up a site for showcasing beauty tips and fashion trends with user-submitted content.
- **Clothing**: Create an online catalogue for a clothing brand, allowing users to filter items by size and style.
- **Gaming**: Build a leaderboard for a gaming community where players can submit their scores and achievements.
- **Food Recipes**: Develop a recipe-sharing website where users can upload, search, and rate recipes.
- **Blogs**: Set up a blogging platform where users can write, edit, and comment on posts.

Write not more than one page outlining the features of your application. Make sure to include the following (Refer to section B for more details):

- Describe the main scope/objective of your web application?
- What technologies will be used to develop the Web Application, in terms of:
    - Database
    - Server-side language
    - CSS framework
    - 3rd party JS framework for client-side validation
- Where will the web application be hosted?
- List the different webpages used to access features of your web application.
- List the user roles in your web application (at least 2)
- List the tables in your database (You may use the phpMyAdmin database designer)
- What data will be added, updated and deleted from the database?

SECTION B – Implementation

The submitted project complies with the proposal as specified in Section A.

The application is to have, at least, the following, and these requirements must make sense in the context in which they are used:

1) Using forms together with client-side and server-side validation
2) Creation of a database with at least 2 tables and a foreign key
3) Connecting with the database (retrieve, update, delete and add)
4) Authentication (Registration and Log in page)
5) Making use of sessions
6) Elimination of duplicate code
7) Interacting with the file system
8) Using a CSS Framework (e.g. Bootstrap 5, Tailwind CSS, Bulma, etc)
9) Hosting online

Criteria and tasks

KU1.2	Identify the components of a web application
Write a minimum of 300 words and upload the proposal document as explained in Section A (**10 marks)**.	

KU1.1	Describe the client/server model with reference to web applications.

Server-side validation to be performed on both the **registration** and **login** forms. Registration form to include email (valid email format), mobile number (not more than 8 numbers) password (min of 5 characters), name and surname should be strings (15 characters each) (**4m**)
Login form to include email and password. All fields are required (**2m**)
The forms to be used in the other web pages (for adding, editing and deleting data) must have 3rd party client side js for their respective validation. Validation libraries to be used (pristine, just-validate etc..) **(4m)**

AA1.3	Compare and contrast static and dynamic web pages

Email and/or mobile number must be unique when creating a new account, that is they don't exist in the database during creation phase. Appropriate messages should be displayed if that is not the case. **(8m)**
Website should be hosted on a web hosting account with full functionality. **(7m)**

AA1.5	Distinguish between stateful and stateless web applications and identify techniques for tackling the stateless web

Sessions to be used properly throughout your web application. **(2m)**
A new session to be created when a user logs in **(2m)**
The session to be destroyed when a user logs out **(2m)**
User has the possibility of logging either with email or mobile **(2m)**
Roles should be applied to logged-in users (the features must be noticeable when compared to anonymous users). **(2m)**

KU2.1	List HTML tags to build static and interactive web pages

Use HTML tags that are essential for collecting user input in forms used in your web application. (**5m**)

Use PHP receive and process this information **(5m)**

KU2.3	Describe how to use CSS selectors to style and format HTML pages, and how to use JavaScript to enhance pages.

Use a CSS framework correctly through-out your whole web application. This includes menus, headers, footers and containers. No broken pages. (**5m**)

Use a CSS framework correctly for your server-side validation (**5m**)

SE3.4	Device code that implements HTTP request handling including sessions, cookies, files and mail.

Profile page
When a user creates and a/c via registration a **profile image** needs to be uploaded. The path of the image to be stored in the database. The image file is stored on the server in a separate folder. **(10m)**
When a user is logged-in the profile image should be shown as a thumbnail, for example in the navbar **(2m)**
The user must be able to change the profile picture once logged-in, with the new image stored successfully on the server. **(8m)**

AA4.2	Use SQL to create and modify databases

Use SQL queries to properly:

- Retrieve from the database and display data as a list on the website **(3 mark)**
- Add data to the list of items on the database **(4 mark)**
- Delete one data item from the database **(3 mark)**
- Update one data item on the database **(5 marks)**

---END OF ASSIGNMENT---